⚽ Football Shot Success Predictor

Fundamentals of AI & ML – BYOP Project Report

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1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are increasingly being used to solve real-world problems across various domains, including sports analytics. In football, decision-making during gameplay is critical, especially when taking shots on goal.

This project focuses on predicting the probability of scoring a goal based on key factors such as distance, angle, and defender pressure. The system uses a machine learning model to analyse patterns and provide predictions along with explanations.

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2. Problem Statement

In real football matches, players often take shots without fully understanding their chances of scoring. Poor decision-making can reduce scoring efficiency.

The objective of this project is to develop an AI-based system that predicts whether a shot will result in a goal or not, and also provides the probability of success. This can help in better understanding shot quality and decision-making.

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3. Objectives

- To build a simple AI model for predicting goal probability
- To apply machine learning concepts learned in the course
- To analyze how different factors affect scoring chances
- To make predictions interpretable using explanation logic

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4. Methodology

4.1 Data Collection

Since a suitable small dataset was not available, a synthetic dataset was created manually. The dataset contains 50+ entries based on realistic football assumptions.

Each entry includes:

- Distance from goal (in meters)
- Angle of the shot (in degrees)
- Defender pressure (0 = low, 1 = medium, 2 = high)
- Goal outcome (1 = goal, 0 = miss)

4.2 Data Preprocessing

- Data was loaded using Pandas
- Features and target variables were separated
- Feature scaling was applied using StandardScaler to normalize input values

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4.3 Model Selection

Logistic Regression was used for this project because:

- The problem is a binary classification problem (goal or no goal)
- It provides probability outputs
- It is simple and easy to interpret

4.4 Model Training

The model was trained using the dataset, where:

- Input features: distance, angle, pressure
- Output: goal (0 or 1)

The model learns patterns from the data and predicts outcomes for new inputs.

4.5 Prediction System

The system takes user input:

- Distance
- Angle
- Pressure

It then:

- Predicts whether the shot will be a goal or miss
- Calculates probability of scoring
- Classifies difficulty level
- Provides explanation based on conditions

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5. Features of the System

- Predicts goal or miss
- Provides probability of scoring
- Classifies difficulty (Easy / Medium / Difficult)
- Gives explanation using rule-based logic
- Visualizes data using graphs

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6. Results and Analysis

The model was able to successfully learn patterns from the dataset.

Observations:

- Shorter distance increases scoring probability
- Better angles improve chances of scoring
- Higher defender pressure reduces success rate

The model achieved good accuracy on training data, indicating that it captured the relationships effectively.

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7. Challenges Faced

- Difficulty in finding a suitable dataset
- Understanding probability outputs from the model
- Handling errors related to libraries and environment setup
- Formatting output for better readability

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8. Limitations

- Dataset is small and manually created
- Model may not perform well on real-world data
- Limited number of features considered

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9. Future Scope

- Use real football match datasets
- Add more features like player skill and shot type
- Improve model using advanced algorithms
- Develop a graphical user interface (GUI)

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10. Conclusion

This project demonstrates how machine learning can be applied to a real-world problem in sports. By using Logistic Regression, the system predicts goal probability and provides insights into shot quality.

The project successfully applies AI and ML concepts such as classification, probability, and model evaluation, making it a practical implementation of course knowledge.

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11. References

- Scikit-learn Documentation
- Python Documentation
- Course Material: Fundamentals of AI & ML

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12. Student Details

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